

MECHANICS OF SOLIDS

STRAIN

01

$$\text{Longitudinal strain} = \frac{\Delta L}{L} = \frac{\text{Change in length}}{\text{Original length}}$$

02

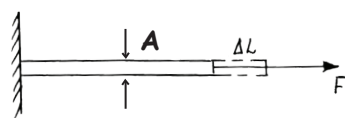
$$\text{Volumetric strain} = \frac{\Delta V}{V} = \frac{\text{Change in volume}}{\text{Original volume}}$$

03

$$\text{Shearing strain} = \phi = \frac{\Delta x}{L}$$

LONGITUDINAL STRESS

Longitudinal/Tensile stress causes increase in length



$$\text{Longitudinal stress} = \frac{F}{A} \text{ unit N/m}^2$$

tensile stress in wire

$$T.S = \frac{F}{A} = \frac{Fm}{(M+m)/A}$$

$$T.S = T = \frac{m(g+a)}{A}$$

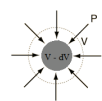
$$T.S_1 = \frac{T_1}{A} = \frac{6mg}{A}$$

$$T.S_2 = \frac{T_2}{A} = \frac{5mg}{A}$$

$$T.S_3 = \frac{T_3}{A} = \frac{3mg}{A}$$

VOLUME STRESS

- Same as pressure
- Causes change in volume



$$\text{volume stress} = \frac{F}{A} = \text{pressure}$$

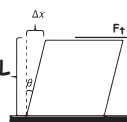
here, F = normal force (or thrust)

SHEARING STRESS

- Causes change in shape

$$\text{shearing stress} = \frac{F_t}{A}$$

F_t = tangential force



HOOKE'S LAW

$$E = \frac{\text{Stress}}{\text{Strain}}$$

(E = modulus of elasticity)

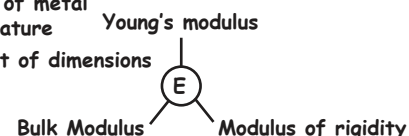
$\tan \theta$ = modulus of elasticity

1. For rigid body E = infinity

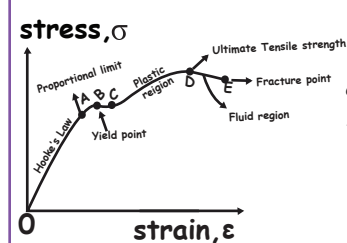
2. Steel is more elastic than rubber

3. Depends on :-
(a) Nature of metal
(b) Temperature

4. Independent of dimensions



STRESS STRAIN CURVE



OA - Hooke's law obeyed

A - Proportional limit

AB - stress is not proportional to strain but body regains its original shape and size regains load is removed

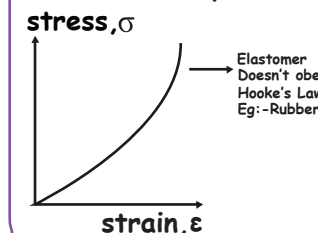
B - Yield point

B to D - Body doesn't regain its original dimension. Beyond B is plastic region.

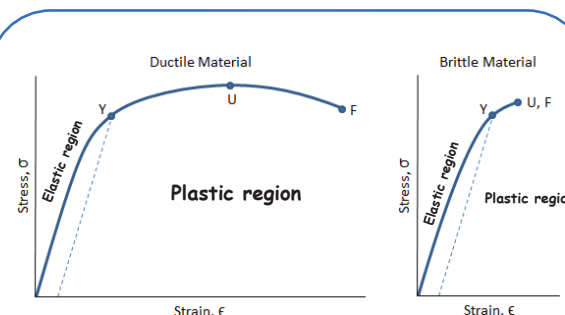
D - Ultimate stress point

Beyond D - large strain is produced even for a small applied force.

E - Fracture occurs



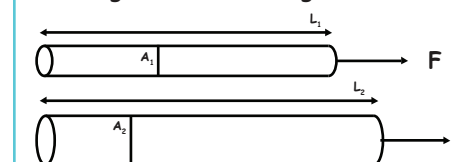
Elastomer Doesn't obey Hooke's Law
Eg:- Rubber



Plastic region is large for ductile materials and smaller for brittle materials

BREAKING STRESS

Breaking Force = breaking stress $\times$ area, $B.F \propto A$



$$\frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow (B.S)_1 = (B.S)_2$$

BREAKING OF WIRE UNDER IT'S OWN WEIGHT

$$B.S \times A = T = mg$$

$$B.S \times A = V \times \rho \times g = A L_{\max} \rho g$$

$$L_{\max} = \frac{B.S}{\rho g}$$

RATIO OF DENSITY OF BODY TO THAT OF LIQUID IN WHICH BODY IS IMMERSED

$$mg = Kl_a$$

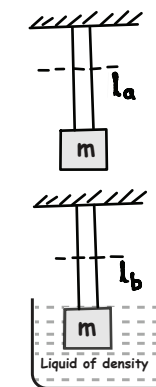
$$mg(1 - \frac{\rho}{\sigma}) = Kl_b$$

$$(1 - \frac{\rho}{\sigma}) = \frac{l_b}{l_a}$$

$$\frac{\rho}{\sigma} = \frac{l_a - l_b}{l_a}$$

$$\frac{\sigma}{\rho} = \frac{l_a}{l_a - l_b}$$

here, l_a and l_b are extensions in the rod in the two cases.



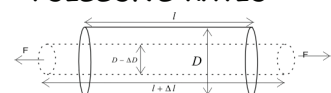
YOUNG'S MODULUS

$$Y = \frac{\text{Longitudinal stress}}{\text{Longitudinal strain}} = \frac{FL}{A\Delta L}$$

Comparing with a spring of force constant K

$$Y = \frac{FL}{Ax} \quad F = \frac{AYx}{L} = kx \quad k = \frac{AY}{L}$$

POISSON'S RATIO

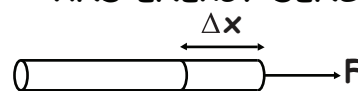


$$\sigma = \frac{\text{Lateral strain}}{\text{Longitudinal strain}} = \frac{(-dr/r)}{(\frac{dl}{l})} = \frac{-\frac{\Delta r}{r}}{\frac{\Delta l}{l}} = -\frac{\Delta D}{\Delta l}$$

Theoretical value :- $-1 < \sigma < 0.5$

Practical value :- $0 < \sigma < 0.5$

ELASTIC POTENTIAL ENERGY AND ENERGY DENSITY



$$E.P.E = \frac{1}{2} kx^2 = \frac{F^2}{2k} = \frac{1}{2} Fx$$

$$= \frac{1}{2} \left(\frac{YA}{L} \right) x^2$$

$$= \frac{1}{2} \times \frac{\text{Stress}}{\text{Strain}} \times \text{volume} \times \text{strain}^2$$

$$= \frac{1}{2} \times \text{Stress} \times \text{strain} \times \text{volume}$$

$$\text{Energy Density} = \frac{E.P.E}{\text{volume}}$$

$$\Rightarrow e = \frac{1}{2} \times \text{Stress} \times \text{strain}$$

INCREASE IN LENGTH DUE TO IT'S OWN WEIGHT

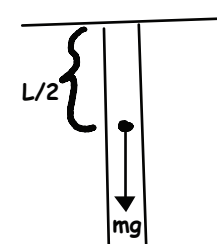
$$F = kx$$

$$mg = \frac{YA}{L} x$$

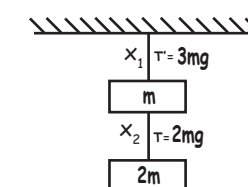
$$x = \frac{mgL}{2YA}$$

$$P.E = \frac{F^2}{2k}$$

$$= \frac{m^2 g^2}{2 \frac{YA}{L}} = \frac{1}{4} \frac{m^2 g^2 L}{YA}$$



RATIO OF EXTENSION



$$\frac{x_1}{x_2} = \frac{\frac{F_1}{k_1}}{\frac{F_2}{k_2}} = \frac{\frac{3mg \times L_1}{Y_1 \times A_1}}{\frac{2mg \times L_2}{Y_2 \times A_2}} = \frac{3l}{2yd^2}$$

$$\text{where } l = \frac{L_1}{L_2} \quad y = \frac{Y_1}{Y_2} \quad d = \frac{D_1}{D_2}$$

BULK MODULUS

$$\text{Bulk modulus, } B = \frac{\text{Volumetric stress}}{\text{Volumetric strain}} = \frac{\Delta P}{-\frac{\Delta V}{V}}$$

$$K = \frac{1}{B} = \text{compressibility}$$

$$B_{\text{isothermal}} = P, \quad B_{\text{adiabatic}} = \gamma P, \quad B_{\text{isotropic}} = \frac{\gamma}{3} P$$

MODULUS OF RIGIDITY

$$\eta = \frac{\text{shear stress}}{\text{shear strain}} = \frac{F}{A\phi} = \frac{Fl}{Ax}$$

